

Table 2. Response quality and preservation of general QA performance

| Method                        | Qwen2.5 7B        |                    |       |       |        |          |            |        |       | GPT-6 Luna        |                   |                   |                   |        |          |            |        |       | Gemma 4 12B       |                    |       |       |        |          |            |        |       | Qwen3.8 27B OFF   |                   |       |       |        |          |            |        |       | Qwen3.8 27B ON    |                   |       |       |        |          |            |        |       |   |   |
|-------------------------------|-------------------|--------------------|-------|-------|--------|----------|------------|--------|-------|-------------------|-------------------|-------------------|-------------------|--------|----------|------------|--------|-------|-------------------|--------------------|-------|-------|--------|----------|------------|--------|-------|-------------------|-------------------|-------|-------|--------|----------|------------|--------|-------|-------------------|-------------------|-------|-------|--------|----------|------------|--------|-------|---|---|
|                               | Cancer-Myth       |                    | CREPE |       | MedQA  | PubMedQA | Medbullets | MMLU   | GSM8K | Cancer-Myth       |                   | CREPE             |                   | MedQA  | PubMedQA | Medbullets | MMLU   | GSM8K | Cancer-Myth       |                    | CREPE |       | MedQA  | PubMedQA | Medbullets | MMLU   | GSM8K | Cancer-Myth       |                   | CREPE |       | MedQA  | PubMedQA | Medbullets | MMLU   | GSM8K | Cancer-Myth       |                   | CREPE |       | MedQA  | PubMedQA | Medbullets | MMLU   | GSM8K |   |   |
|                               | FPQ ↑             | NFP ↑              | FPQ ↑ | TPQ ↑ | Acc. ↑ | Acc. ↑   | Acc. ↑     | Acc. ↑ | EM ↑  | FPQ ↑             | NFP ↑             | FPQ ↑             | TPQ ↑             | Acc. ↑ | Acc. ↑   | Acc. ↑     | Acc. ↑ | EM ↑  | FPQ ↑             | NFP ↑              | FPQ ↑ | TPQ ↑ | Acc. ↑ | Acc. ↑   | Acc. ↑     | Acc. ↑ | EM ↑  | FPQ ↑             | NFP ↑             | FPQ ↑ | TPQ ↑ | Acc. ↑ | Acc. ↑   | Acc. ↑     | Acc. ↑ | EM ↑  | FPQ ↑             | NFP ↑             | FPQ ↑ | TPQ ↑ | Acc. ↑ | Acc. ↑   | Acc. ↑     | Acc. ↑ | EM ↑  |   |   |
| TF-IDF text gate              | 50.7 <sup>H</sup> | 60.4 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 63.8 <sup>H</sup> | 50.3 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | 68.3 <sup>H</sup> | 49.0 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 78.2 <sup>H</sup> | 48.3 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| Plain                         | 2.4 <sup>H</sup>  | 100.0 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 54.5 <sup>L</sup> | 91.0 <sup>L</sup> | 72.7 <sup>R</sup> | 98.9 <sup>R</sup> | —      | —        | —          | —      | —     | 4.3 <sup>H</sup>  | 100.0 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 49.7 <sup>H</sup> | 79.9 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 40.7 <sup>H</sup> | 88.6 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| CoT                           | 0.7 <sup>H</sup>  | 99.3 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 3.4 <sup>H</sup>  | 99.3 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | 54.7 <sup>H</sup> | 71.1 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 43.1 <sup>H</sup> | 87.9 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| Balanced instruction          | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — |   |
| Premise-review CoT (2-step)   | 10.5 <sup>H</sup> | 94.6 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 41.2 <sup>H</sup> | 78.5 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | 61.4 <sup>H</sup> | 55.0 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 66.7 <sup>H</sup> | 57.0 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| Always correct                | 56.9 <sup>H</sup> | 21.5 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 72.0 <sup>H</sup> | 2.0 <sup>H</sup>   | —     | —     | —      | —        | —          | —      | —     | 71.0 <sup>H</sup> | 28.9 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 83.4 <sup>H</sup> | 5.4 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | — | — |
| PreWoMe-style                 | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | 88.9 <sup>L</sup> | 59.0 <sup>L</sup> | —                 | —                 | —      | —        | —          | —      | —     | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — |   |
| Extract + Verify              | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | 87.9 <sup>L</sup> | 48.0 <sup>L</sup> | —                 | —                 | —      | —        | —          | —      | —     | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — |   |
| Direct-gated response         | 6.9 <sup>H</sup>  | 90.6 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 37.2 <sup>H</sup> | 83.9 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | 53.0 <sup>H</sup> | 72.5 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 53.5 <sup>H</sup> | 75.8 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| Premise-review gated response | 21.3 <sup>H</sup> | 85.9 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | 43.4 <sup>H</sup> | 67.1 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | 62.1 <sup>H</sup> | 59.7 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | 62.1 <sup>H</sup> | 60.4 <sup>H</sup> | —     | —     | —      | —        | —          | —      | —     | — | — |
| Probe-gated response          | 44.8 <sup>H</sup> | 64.4 <sup>H</sup>  | —     | —     | —      | —        | —          | —      | —     | n/a               | n/a               | n/a               | n/a               | n/a    | n/a      | n/a        | n/a    | n/a   | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — | — |
| GEPA                          | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | 78.8 <sup>L</sup> | 72.0 <sup>L</sup> | 74.7 <sup>R</sup> | 98.0 <sup>R</sup> | —      | —        | —          | —      | —     | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — | — |
| Ours                          | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —                 | —                 | —      | —        | —          | —      | —     | —                 | —                  | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | —                 | —                 | —     | —     | —      | —        | —          | —      | —     | — | — |

Stored results; —: not measured/comparable; n/a: outside the defined implementation. Percentages, one run.  
H: historical full-corpus Cancer-Myth; valid denominators and protocol differences in ledger.  
FPQ / NFP / TPQ: Well  $\geq 4$ . Acc.: accuracy; EM: exact match.  
L: Luna matched 99 FPQ / 100 NFP, exposed test; R: CREPE 751 FPQ / 2253 TPQ.  
Gate rows route saved Plain/correction answers; not exact Two Axes reproduction.  
Ours is not fixed to SAE. Source counts and comparison limits: result\_sources.md.  
Each model contains every dataset; QA entries remain unmeasured (—), not zero.